Course milestone M8: Robustness Audit

HONR 46400 · Evidence-Driven Research

What to submit

One document, `lastname_m08.pdf`. Everything this milestone asks for goes inside it, as sections in the order below. One artifact travels beside it, because it cannot live inside a PDF:

Submit it on Brightspace, under **Assignments** then **M08**. Brightspace carries the deadline.

This milestone presents **Book Milestone 8: Your robustness audit** (checkpoint, version 1) of the course book, EDR|AI.

- `Shared, rerunnable Colab notebook link (the audit notebook)`: The brief requires the instructor to open AND rerun the audit so every reported check reproduces. A PDF cannot execute code, so the live notebook has to be shared alongside it; the link is pasted into the PDF but the runnable object itself is separate.

What this course adds

Freeze your three checks at the week's lecture

1. Your pre-list is exactly three checks, and each one names the reviewer attack it answers.
2. At the transfer block that closes the week's lecture, put each check into the studio workbook as the exact code you will run, in quotes.
3. The workbook parses that code for syntax and runs nothing else. What leaves the room is a commitment, not a result.
4. Run none of the three before the Friday studio. A check that occurs to you once numbers are on screen is still worth running: label it exploratory and keep it off the frozen list.

Run and adjudicate at the Friday studio

You execute the frozen audit at the studio, working with your AI assistant.

1. The adversarial review has a named reviewer here: the **Robustness & Sensitivity Reviewer** (briefing: `genai_studio/roles/robustness_sensitivity_reviewer.md`).
2. One studio does not hold enough minutes for a check per flag, so you triage. Rank the flags by how much damage each would do to your claim if it were true.
3. Verify the most damaging flag in the room. Record four fields: the flag as raised, the check you ran, the output it produced, and your verdict of confirmed or refuted.
4. Put every remaining flag on a **pending register**: the flag, its damage rating, the check that would settle it, and when you will run it. Pending is an honest state here, and it is graded as one.
5. Where one of your checks refuted a flag, name the **loudest wrong flag**: the objection the reviewer stated most confidently that your data did not support.
6. Walk the room through what survived, defend the flag you verified there, and hand over your register. Submit the same day.

Hand over a notebook that reruns

Your audit runs in a Colab notebook. Set sharing so the instructor can open **and rerun** it, and paste that link inside your PDF. Every check you report has to run from that notebook, so restart it and confirm the audit reproduces before you submit.

How this part is judged

These course pieces are read inside the Book Milestone rubric. Graders look for a scaffold frozen at the lecture, the top flag settled by a check whose output you show, a complete register for the rest, and a notebook that reruns.

Milestone 8: Your robustness audit

Open the milestone workbook in Colab

This chapter closes Studio 8: Stress-test and adjudicate. Its lessons each ended at **It is your turn**; what you wrote there arrives here as working material, and this is where the pieces become one artifact you can defend.

What this milestone produces. A pre-listed robustness grid, negative tests with their assumptions stated, diagnostics, an adversarial-review record, and your adjudication of what survived.

What you bring

You also carry forward the last milestone's artifact: **Milestone 7: Your first reproducible analysis**. This one builds on it, and the chain assumes it is versioned and filed.

Check you are carrying each of these before you start. If one is missing, go back and work that lesson's **It is your turn** first; the practice below assumes it.

- Lesson 24: Robustness and Sensitivity: your pre-listed robustness grid with its full results, direction and size, plus the named null check.
- Lesson 25: Diagnostics and Negative Tests: your negative test and diagnostic, each with the prediction you wrote before running it.
- Lesson 26: AI as Adversarial Reviewer: your commissioned adversarial review with every flag adjudicated against the data, confirmed or refuted.
- Lesson 27: Recognizing False Confidence: your confidence audit: the not-yet-recomputed list, where belief in the headline number comes from, and your boundary sentence.

The practice

1. Pre-list your checks before you run any of them, and commit to reporting all of them.
2. Run the robustness grid and report a span within each commensurable panel, never across panels.
3. Run the null check that your design licenses, and name it correctly.
4. Commission at least one adversarial review, and add a human reviewer when one is available. Neither review is independent verification: confirm or refute every flag with a data check.
5. Write the adjudication: what survived, what did not, and what you still cannot rule out.

A version, not a pass

Your milestone artifact is a dated, numbered **version** with the reason for the version attached. When later evidence changes it, you write the next version rather than editing the last one, because the sequence of changes is itself part of your research record.

The four rails, here

Four concerns cross every studio in this book. At this milestone they take these specific forms.

- **Ethics, permissions, and data exposure.** Report the checks that hurt your finding as fully as the ones that helped.
- **Evidence, provenance, and reproducibility.** A flag is real when a check confirms it, not when the reviewer sounds certain.
- **AI activity, verification, and human decisions.** An AI reviewer is another critique, not independent verification.
- **Uncertainty, claim boundary, and revision history.** Specification spread measures your choices; it is not an uncertainty interval.

Where this milestone sits

You arrived carrying Milestone 7: Your first reproducible analysis. Milestone 9 writes down only the claims this audit licenses, at the strength it licenses them. The chain ends at Milestone 12: Your release and next cycle, where you decide whether your finished research artifact leaves your hands.

What sends you back

A finding that does not survive its own pre-listed checks returns to Studio 5 or 7, not to the writing studio.

How this milestone is assessed

Each row scores **0**, **1**, or **2**. **10 points total.**

#	Criterion	0	1	2
1	The milestone artifact exists in full, specific to your own project	absent, or generic text that would fit any project	present but incomplete, or specific in some parts and generic in others	complete and unmistakably about your project
2	The milestone is written as a dated, numbered version with its reason	no version, or a version with no reason given	versioned, but the reason is decorative rather than an account of what changed	versioned with a reason a reader could use to reconstruct your thinking
3	Every judgment in the artifact is defended by you, not asserted by a tool	conclusions appear with no reasoning, or reproduce a tool's wording	reasoning present but thin where the decision was hardest	each judgment carries a reason you could defend under questioning
4	The four rails are carried in the form this studio requires	one or more rails absent where this studio required them	all rails present but one is nominal	every rail addressed in the form this studio requires
5	The milestone's own product is present and defensible: A pre-listed robustness grid, negative tests with their assumptions stated, diagnostics, an adversarial-review record, and your adjudication of what survived	the milestone's own product is missing	present but would not survive a careful question	present and defensible as stated

! Blocking gate

No work at or after Studio 4 proceeds past a not-authorized determination. This is not scored and cannot be averaged away.

Your workbook

Open the workbook with the link above. It walks these steps with a cell for each one, ends by writing your milestone version with its reason, and adds the rows this studio contributes to your AI Research Ledger and your Research Dossier.